

Data Cleaning & Preparation Report

Project Title

Data Cleaning and Preparation

Intern Name

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Internship

DecodeLabs – Data Analytics Industrial Training Kit (Project 1)

Objective

The objective of this project was to clean a raw dataset by identifying and handling missing values, checking for duplicate records, correcting data formats, and ensuring the dataset was accurate and ready for further analysis.

Dataset Overview

- Total Records: **1,200**
- Dataset Type: Sales/Order Data
- Tool Used: **Microsoft Excel**

Tasks Performed

1. Missing Value Analysis

The dataset was examined for missing or blank values using Excel's **Go To Special → Blanks** feature.

Findings:

- A total of **309 blank cells** were found in the **CouponCode** column.
- No missing values were found in any other column.

Action Taken:

- All blank values in the **CouponCode** column were replaced with **"No Coupon"** to indicate that no discount coupon was applied.

2. Duplicate Record Check

The **OrderID** column was checked for duplicate records using Excel's **Conditional Formatting (Duplicate Values)** feature.

Findings:

- No duplicate Order IDs were found.
 - All Order IDs were unique.
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3. Data Format Validation

The dataset was reviewed to ensure data consistency.

The following checks were performed:

- Date values were formatted consistently.
 - Numeric columns (Quantity, Unit Price, Items in Cart, and Total Price) were verified as numeric values.
 - Text fields were checked for consistency.
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4. Data Verification

To verify the accuracy of the dataset, the **Total Price** column was validated using the following calculation:

Formula Used:

Total Price = Quantity × Unit Price

The calculated values matched the existing **Total Price** values, confirming that the dataset calculations were correct.

Results

Task	Status
Missing Values Identified	Completed
Missing Values Handled	Completed
Duplicate Order IDs Checked	Completed
Date Format Verified	Completed

Task	Status
Numeric Data Verified	Completed
Total Price Validation	Completed

Conclusion

The dataset was successfully cleaned and prepared for analysis. Missing values were handled appropriately, duplicate records were checked, data formats were validated, and calculated values were verified. The cleaned dataset is now accurate, consistent, and ready for further data analysis and visualization.